

BORDER, QC, Canada

WMO: 719010

Lat: 55.333N

Lon: 63.202W

Elev: 479

StdP: 95.70

Time Zone: -4.00 (NAA)

Period: 86-98

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-35.2	-33.9	-39.8	0.1	-35.4	-38.0	0.1	-33.7	21.4	-20.8	18.8	-20.8	4.7	270	0.952

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.4	21.9	13.7	19.8	12.8	17.2	11.5	15.0	19.3	13.5	17.0	12.3	15.8	5.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.3	10.1	15.9	11.9	9.2	14.6	10.5	8.4	13.7	43.7	19.0	39.5	17.3	36.3	15.7	18.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.5	13.9	12.0	-39.2	26.1	1.2	3.2	-40.1	28.4	-40.8	30.3	-41.4	32.1	-42.3	34.5		
(5)			-38.9	17.3	1.8	1.1	-40.2	18.1	-41.2	18.7	-42.2	19.3	-43.5	20.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-5.3	-23.2	-23.2	-16.0	-7.2	0.1	6.4	10.7	10.1	5.1	-1.0	-8.5	-17.8
	DBStd	13.11	7.41	6.86	7.09	5.71	3.92	4.51	3.46	3.74	3.22	3.36	5.85	7.67
	HDD10.0	5708	1029	931	807	517	308	127	34	46	151	342	554	862
	HDD18.3	8626	1288	1164	1066	767	566	359	238	257	397	601	804	1120
	CDD10.0	127	0	0	0	0	1	19	55	48	4	0	0	0
	CDD18.3	2	0	0	0	0	0	2	0	1	0	0	0	0
	CDH23.3	39	0	0	0	0	0	24	4	11	0	0	0	0
	CDH26.7	11	0	0	0	0	0	10	0	1	0	0	0	0
	WSAvg	5.0	5.4	4.9	5.5	4.7	4.2	4.7	4.3	4.7	5.1	5.2	5.8	5.4
(15) Precipitation	PrecAvg	753	51	43	44	46	49	71	110	81	84	72	56	46
	PrecMax	981	104	102	113	119	138	110	185	140	172	133	97	143
	PrecMin	516	4	2	11	8	9	20	47	43	27	40	10	8
	PrecStd	103	24	27	27	24	25	20	31	22	31	23	20	30
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.1	-1.1	1.2	5.2	15.0	27.0	24.0	24.9	16.9	8.9	5.8	0.2
		MCWB	1.9	-1.3	-0.3	2.1	8.8	16.3	15.5	16.1	10.4	7.0	5.7	0.0
	2%	DB	-3.1	-4.1	0.1	3.1	10.8	20.2	21.8	21.2	14.2	6.8	2.1	-1.0
		MCWB	-3.5	-4.4	-0.8	0.9	5.8	13.1	13.5	14.1	10.5	4.9	1.5	-1.2
	5%	DB	-6.2	-7.8	-1.2	2.0	7.9	17.1	19.8	19.0	12.1	5.0	0.2	-3.8
		MCWB	-6.5	-8.4	-1.7	0.2	4.1	11.3	12.4	12.6	9.2	3.5	-0.5	-4.3
	10%	DB	-10.9	-11.8	-4.1	0.9	5.2	14.2	17.8	16.8	10.2	3.1	-0.8	-6.1
		MCWB	-11.4	-12.3	-4.9	-0.4	2.6	9.6	11.5	11.7	8.1	1.9	-1.6	-6.6
	0.4%	WB	0.2	-1.6	0.2	2.7	8.9	16.4	16.2	17.2	12.7	7.6	5.3	-0.3
		MCDB	0.9	-1.4	0.6	4.4	14.6	21.0	20.1	21.6	14.8	8.5	5.4	-0.2
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-3.7	-4.6	-0.7	1.5	6.1	13.7	14.5	15.1	10.9	5.3	1.5	-1.5
		MCDB	-3.4	-4.3	0.0	2.8	9.6	18.0	18.0	19.3	12.9	6.2	1.9	-1.2
	5%	WB	-7.0	-8.3	-2.3	0.4	4.4	11.9	13.4	13.5	9.5	3.5	-0.3	-4.2
		MCDB	-6.7	-7.8	-1.6	1.4	7.0	15.6	16.8	17.2	11.5	4.6	0.1	-3.6
	10%	WB	-11.5	-12.3	-5.1	-0.4	3.0	10.1	12.2	12.3	8.3	2.0	-1.5	-6.6
		MCDB	-11.1	-11.8	-4.4	0.6	5.0	13.6	15.9	15.5	10.4	3.0	-0.8	-6.1
(35) Mean Daily Temperature Range	5% DB	MDBR	8.4	9.3	10.6	8.9	7.6	9.2	9.4	8.7	6.3	4.5	6.1	8.0
		MCDBR	13.0	11.6	10.3	9.1	11.6	15.2	13.7	12.8	9.6	6.6	6.3	8.0
		MCWBR	13.0	11.6	9.8	7.4	7.2	7.8	6.0	6.0	6.4	5.3	6.5	8.0
	5% WB	MCDBR	13.1	11.7	10.7	7.7	10.2	13.3	10.4	10.5	9.0	6.0	6.3	8.1
		MCWBR	13.0	11.6	10.4	6.6	6.9	8.0	6.3	6.3	6.9	5.4	6.7	8.2
(40) Clear-Sky Solar Irradiance	taub		0.227	0.242	0.268	0.332	0.392	0.368	0.378	0.354	0.329	0.284	0.240	0.222
	taud		2.205	2.251	2.202	2.062	2.044	2.335	2.361	2.426	2.462	2.462	2.251	2.119
	Ebn at Noon		728	858	909	883	845	876	859	861	838	801	700	625
	Edh at Noon		55	78	105	142	154	117	112	99	83	64	52	47
(44) All-Sky Solar Radiation	RadAvg		0.89	1.96	3.57	5.04	5.47	5.20	4.60	3.78	2.73	1.41	0.65	0.57
	RadStd		0.07	0.18	0.32	0.38	0.54	0.47	0.33	0.35	0.23	0.12	0.11	0.07

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page